

INFRARED SMALL MOVING TARGET DETECTION BASED ON SPATIAL-TEMPORAL-FREQUENCY ANALYSIS

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ABSTRACT

The Infrared Searching and Tracking System (IRSTS) plays an important role in both military and citizen life, contributing to crucial areas such as security surveillance and transportation, and so on. As the core step in the system, methods for infrared moving target detection and tracking attract much attention in recent years, especially for several specific small targets like unmanned aerial vehicles (UAVs). In this work, a spatial-temporal-frequency feature fusion based approach is proposed. Firstly, the infrared sequential images are decomposed via 3D wavelet transform. Only the high frequency information in spatial and temporal domains is reconstructed to form the reconstructed sequence. A background suppression strategy from temporal to spatial domain is secondly applied in the reconstruction images and the target is acquired by adjust segmentation. Experimental results demonstrate that the proposed method achieves competitive performance both in qualitative and quantitative comparisons.

Index Terms— infrared searching and tracking, infrared small moving target detection, 3D wavelet transform, spatial-temporal-frequency analysis

1. INTRODUCTION

Infrared searching and tracking systems (IRSTS) is developing rapidly by virtue of all-day monitoring and strong anti-interference ability of infrared imaging[1]. Due to the long imaging distance and the diverse nature of the backgrounds, detecting moving targets is a challenging and key procedure in IRSTS. Infrared small moving target detection methods have been studied a lot in the past few years.

The mainstream approaches for infrared small target detection are network based methods and traditional detection methods. Data-driven and deep learning models that have been proven effective in other computer vision tasks[1], including generative adversarial network (GAN)[2, 3] have been successfully applied to infrared small target detection. Furthermore, the characteristics of infrared small target

such as shape and local contrast feature are also combined with attention mechanism and context modulation in the networks[4, 5, 6, 7]. However, the deep learning approaches heavily relied on datasets[5], image qualities[8] and spatial information. Therefore, filtering methods, saliency features, low rank and sparse decomposition (LRSD) remain popular for infrared small target detection. Specifically, the filter-based methods are easy to operate and can acquire good results in simple scenes with high time efficiency. When backgrounds are complex, filter-based methods suffer from high false alarm rate. Thus, the combination of several filters are built to describe the saliency feature, local contrast measure (LCM)[9], of infrared small target, inspired by human visual system (HVS). Although many techniques such as weighting, multi scale, double/tri-layer window which performance well are applied to LCM[10, 11, 12], these approaches primarily depend on spatial features with minor utilization of temporal information. Similarly, LRSD methods have been developed from infrared patch image (IPI) model[13] to infrared patch tensor (IPT) model[14] along with information of spatial domain to spatial-temporal domain and achieve superior performance for complex backgrounds[15, 16, 17] while high computation complexity cannot be avoided.

Despite these methods, infrared small moving target detection still face the challenges of lack of features, low-quality images and heavy interference. The majority of existing algorithms solely use spatial features or spatial-temporal features[18], leading to a deficiency in frequency information. However, the moving target is the high frequency component both in spatial axes, horizontal and vertical, as well as in the third dimension, temporal axis. Therefore, in this work, we employ 3D wavelet decomposition and reconstruction to intuitively extract high-frequency components in three dimensions for target detection. After the frequency analysis, we use the mean value in a time period to adjust the images in the corresponding period which thus combine the spatial and temporal information to suppress the residuals of reconstructed sequences and enhance the targets. The

contributions of this letter mainly consist of 2 aspects: the utilization of frequency characteristics and the spatial-temporal enhancement method which means the spatial, temporal and frequency domain are employed at the same time.

2. METHODOLOGY

In this letter, a novel infrared small moving target detection method based on spatial-temporal-frequency analysis is proposed. The proposed method uses: 1) 3D wavelet decomposition and reconstruction; and 2) spatial-temporal adjustment. This approach can be described in 4 steps: 1) 3D wavelet transform of infrared sequence images; 2) set other sub-bands to zero except the high frequency sub-band in three dimensions and reconstruct the volume; 3) for each frame in the sequence, find the temporal mean value of images in a time period and adjust the current frame; 4) threshold the image and extract the target. The flowchart of the proposed algorithm is shown in Fig 1 and the details are presented in the following.

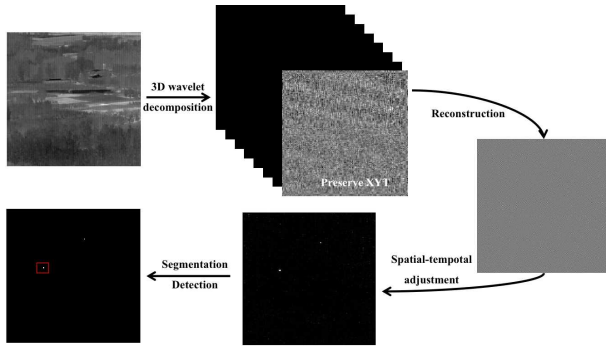


Fig. 1. Flowchart of the proposed detection method.

2.1. 3D wavelet decomposition and reconstruction

In order to suppress the background and enhance the target at the same time, the correct frequency components should be preserved and others should be inhibited. Through the analysis of frequency domain, we observe that the low frequency components in spatial and time domains represents the static background and high frequency components in either spatial vertical or horizontal axes describe the different directions of edges and temporal stationary interference. Therefore, after the decomposition, all of sub-bands should be suppressed except for the high frequency component in three dimensions.

In this work, we use lowercase and capital to represent the low frequency component and high frequency component respectively. The infrared sequence images are firstly decomposed by 3D wavelet transform[19, 20] and 8 sub-bands are acquired. Then, the sub-bands with low frequency information (xyt, Xyt, xYt, xyT, XyT, xYT, XYt) are set to zeros

while the other sub-band (XYT) is preserved. Consequently, inverse transform is operated and thus the infrared sequence images are reconstructed with the suppression of non-target compositions.

2.2. Spatial-temporal adjustment and target detection

After the analysis in frequency domain, there are still some residuals such as nonstatic isolated noise and moving backgrounds. To address the problem of these interference, we sample the previous and later N frames for each image in the reconstructed sequences. Then, the mean background for these frames is calculated using Equation 1, which is the mean value of each pixel.

$$\mu BKG_i = \text{mean}(R_{i-N}, \dots, R_i, \dots, R_{i+N}) \quad (1)$$

where μBKG_i represents the mean background of each i frame in the reconstructed sequences and R_i means the i^{th} reconstructed frame. Subsequently, the corresponding frame should remove its mean background to eliminate the interference. To avoid the suppression of the target simultaneously, we propose to enhance the target via Equation 2.

$$F_i = (R_i - \mu BKG_i)^2 \quad (2)$$

Here, F_i is the result of spatial-temporal adjustment. This computation method in a period time can avoid the fierce changes in the whole sequence, decreasing the false alarm rate to some extent and enhance the real target as well. Finally, the target is detected by the adjust threshold segmentation.

3. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, the proposed algorithm is tested on 4 sequences and compared with other 4 methods. The parameter settings is listed in Table 1. All the parameters of the methods are consistent with the parameter settings in the corresponding literature of the methods.

Fig 2 demonstrates the detection results of different methods for 4 sequences with targets marked in red and false alarms marked in blue and we can see that the proposed approach extract the target automatically and precisely. Moreover, our algorithm achieves competitive performance compared with other existing methods.

Furthermore, quantitative experiments are conducted. Similar to [16], the background suppression factor (BSF), signal-to-clutter ratio gain (SCRG), receiver operating characteristic curves (ROC) and the area under curve (AUC) are considered. The quantitative performance is illustrated in Table 2 and the best results are marked in red which demonstrates that the proposed algorithm in this letter acquires the superior or the sub-optimal performance. The corresponding ROC is also depicted in Fig 3, where q is the threshold value. From Fig 3 we can also conclude that the proposed method

Table 1. Parameter settings of different methods

Method	Parameters
STLDM[18]	frames=5
PSTNN[21]	patch size: 40×40 , step: 40, $\lambda = 0.7 * n3$, $\varepsilon = 10^{-7}$
TLLCM[22]	Gaussian filter kernel
WTLLCM[12]	window size: 3×3 , $K=4$
Proposed	decomposition level=1, wavelet function: biorthogonal, $N=5$

Table 2. Performance of different methods

Methods	Seq1			Seq2			Seq3			Seq4		
	BSF	SCRG	AUC	BSF	SCRG	AUC	BSF	SCRG	AUC	BSF	SCRG	AUC
STLDM	3.852	2.862	0.7749	7.568	3.443	0.9108	5.953	3.153	0.8979	6.273	6.266	0.7578
PSTNN	24.828	2.12	0.8967	48.434	3.083	0.9332	23.693	1.939	0.8607	48.043	5.212	0.8416
TLLCM	18.859	1.902	0.9057	28.222	2.474	0.9299	25.803	2.433	0.9063	13.935	0.902	0.0679
WTLLCM	74.332	1.968	0.9873	213.543	2.337	0.9495	79.044	2.534	0.9489	16.601	3.144	0.3164
Proposed	68.407	23.01	0.9949	225.724	6.991	0.9541	156.949	8.948	0.9581	50.952	20.17	0.9222

can achieve the competitive performance compared with the existing approaches. This is due to its integration of spatial-temporal-frequency analysis and 3D wavelet decomposition and reconstruction, which effectively suppresses non-target compositions and enhances the target detection accuracy. Additionally, the spatial-temporal enhancement strategy mitigates non-static interference, leading to superior performance in both precision and robustness.

In addition to the detection results, time efficiency is important for infrared small moving target detection as well. Therefore, the time cost is computed and shown in Table 3. Actually, the speed still needs to be improved in the future work although we have already acquire superior speed.

Table 3. Time of different methods (in seconds)

Methods	Seq1	Seq2	Seq3	Seq4
STLDM	1.8672	1.8553	1.7634	1.7867
PSTNN	0.2299	0.1978	0.2337	0.1661
TLLCM	1.4250	1.4751	1.3667	1.1157
WTLLCM	0.2781	0.2614	0.3025	0.2933
Proposed	0.0145	0.0138	0.0146	0.0141

4. CONCLUSIONS

This letter proposes an infrared small moving target detection method based on spatial-temporal-frequency analysis with the concept of 3D wavelet decomposition and reconstruction. The spatial-temporal enhancement is also proposed to avoid the non-static interference. The qualitative and quanti-

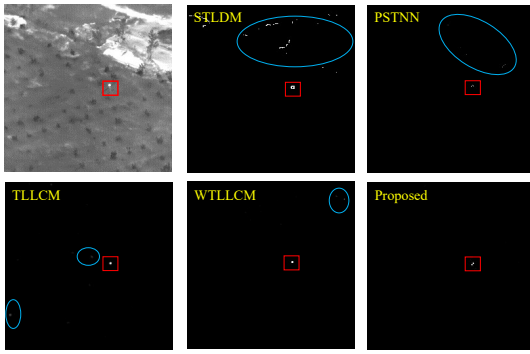
tative experiments show that the proposed algorithm achieves the superior or competitive performance compared with other methods which means the proposed work can detect the target precisely and fast.

5. ACKNOWLEDGEMENTS

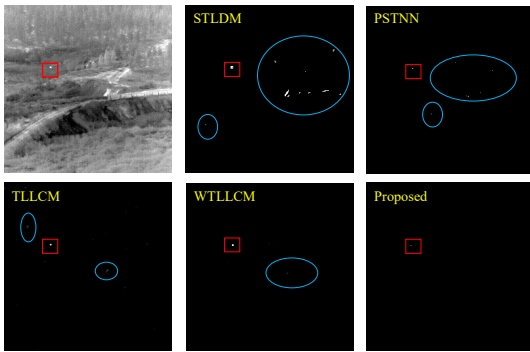
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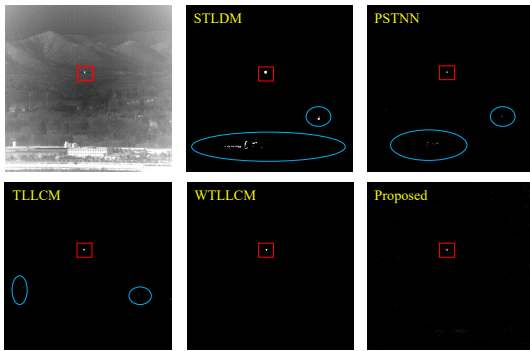
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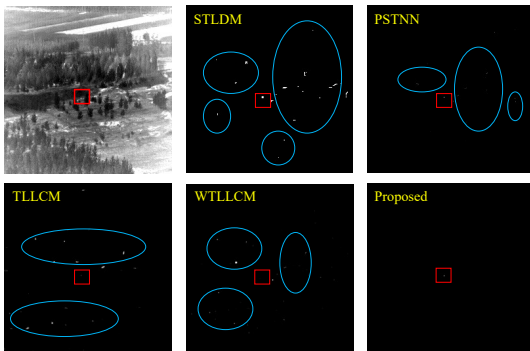
(a)



(b)

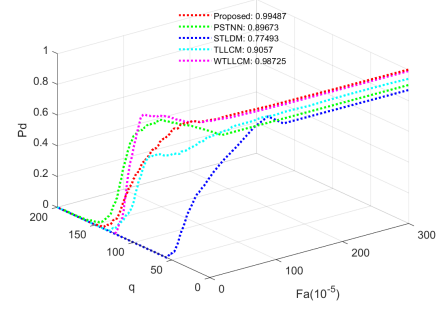


(c)

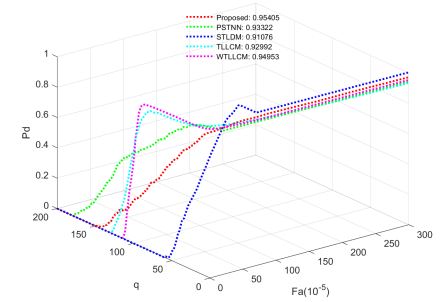


(d)

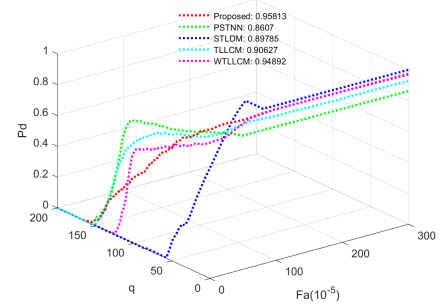
Fig. 2. Detection results of different methods. Red box means the target while blue ellipses are the false alarms.



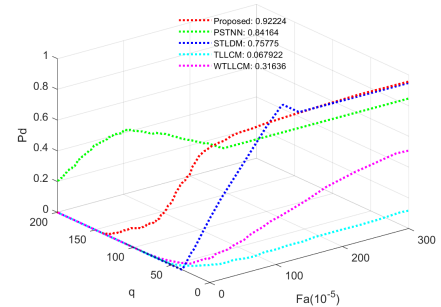
(a)



(b)



(c)



(d)

Fig. 3. ROC of different methods for 4 sequences.

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